

SIG: Grammar Lexicons and Wordnets

Scribes: Dan and Luis

Luis: Interest in DELPH-IN + Wordnet alignments for Cantonese, Mandarin, Norwegian, in addition to English.

Luis: One can have the lexical types split by synsets and then ambiguated into all possible types/subcats that are possible for that synset;

This can then maybe be useful because you can reduce this back into the old world, but also allows you to merge synsets into any other merge strategy

- parse selection (potential, unclear)
- making ambiguity visible (clear, potentially risky for other thing)
- Dan is interested into human/non-human distinction (suggested as)
- the construction "the doctor's office / every doctor's office / ..."
- Some grammar checking (e.g. who vs which) you also need this
- anaphoric binding is also possibly interesting to have this information
- In the service of people who want to do reasoning downstream; the grammar already makes sense distinctions that are syntactically motivated; if a user has to word sense disambiguate by themselves, they lose the information that is available in the grammar (e.g. subcategorization)

Dan: how can I exploit (for better parse ranking) semantic information like "with binoculars" which is used to modify humans and not very often with animals and never with trees and plants etc.

I would need to make it a property that is visible in the semantic index, or find some other way to expose it to inform parse ranking.

Grandparenting in training a parse selection model currently goes up to 2 with the ERG; this is 1 level short for all of these;

EB: there was something related with head projection in earlier discussions

Dan: Don't remember anything like this ever being actually implemented.

EB: In computing these subtrees, we might try appending the Wordnet synset value at the end of the rule name; you create those things by pulling them out of the feature; would proliferate names, and would need to be done on the fly in parsing – not obvious how.

EB: we should also thinking about using information that is provided in previous context (incl. previous sentences?) to rank parses

EB: also, could you use the idea of ubertagging to compress the lexical rule unary chains in the tree and get more visibility even with current grandparenting of only 2?
John: mentioned also about reducing lexical and unary rules

Dan: Consider strategy of re-ranking, first using the existing syntactic information in the present parse selection model, then producing some number of top-ranked trees and use semantic information to re-rank them in order to find a good tree that is also consistent with semantic type constraints derived from Wordnet.